

Benchmark design governs reported skill in daily river water-temperature prediction

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Key points

- Confirmatory: seven-day skill of +0.250 against persistence falls to +0.038 against damped persistence on identical keys.
- Descriptive: withholding a gauge's own recent water temperature from the inputs costs 1.3-1.7 °C; the model class costs under 0.01 °C.
- Descriptive: future weather is worth 0.13-0.63 °C, and worth less once the local gauge is gone, so the two are complements.

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Abstract

Daily river water temperature is strongly persistent and seasonal, so reported machine-learning skill depends on what a model is compared against and on what it may see. On one frozen panel of 120 U.S. gauges across 15 hydrologic regions we vary the evaluation design and the information supplied, scoring every arm on one common registry of station/date/horizon keys with all preprocessing fitted strictly backwards in time. On a pre-specified 2021-2023 held-out window, a constrained deep predictor's seven-day station-median skill is +0.250 against persistence but +0.038 against damped persistence, and a gradient-boosted tree with site identity is most accurate at one and three days. All five sealed tests are supported at their registered margins, two of them non-inferiority tests against a pre-declared +0.05 °C ceiling that carries no ecological interpretation. Descriptive analyses on the same keys measure what the information is worth. Withholding a gauge's own recent water temperature from the inputs costs 1.3-1.7 °C at every lead and at all 116 stations, two orders of magnitude more than the gap between the two estimator implementations compared. Realized future meteorology is worth 0.13-0.63 °C, is almost entirely future air temperature, and survives two pre-registered placebos. Future weather and local observations are complements rather than substitutes: forcing is worth less once those observations are withheld, and the estimator begins to matter only where information is scarce. The held-out comparison is confirmatory; every information result is post-outcome and descriptive, each an input ablation at gauged sites, not prediction where no record exists.

Plain Language Summary

River temperature changes slowly from day to day, so a forecast that repeats yesterday's reading is already fairly accurate, and nudging it toward the season's usual value is better still. We tested predictions at 120 U.S. gauges under one demanding evaluation: every model scored on exactly the same days and sites, none allowed to see anything dated after its issue time. What a model was given mattered far more than which method it used. Denying a model the recent readings from the gauge it is predicting cost roughly two hundred times more accuracy than switching between entirely different families of model. Letting it see the weather that has not happened yet was worth about a third as much, came almost entirely from air temperature, and — unexpectedly — was worth less, not more, once those readings were gone. These are comparisons between what a model is fed,

at rivers that do have thermometers; none of them tests where a thermometer should be installed. Much of what is usually reported as a modelling advance is, on this evidence, a statement about what the model was given and what it was compared against.

Keywords: river water temperature; benchmark design; strong baselines; spatial holdout; comparative evaluation; information value.

1 Introduction

Stream thermal regimes set dissolved-oxygen saturation, metabolic rates, life-stage timing, and habitat suitability, and respond to atmospheric forcing, discharge, groundwater exchange, shading, channel geometry, and regulation (Caissie, 2006). Daily records from national monitoring networks have made the variable one of the most heavily modeled targets in applied hydrological machine learning, and the prevailing account is that deep sequence models have substantially advanced it: non-linear air–water regressions (Mohseni et al., 1998) and hybrid air-temperature and discharge formulations (Toffolon & Piccolroaz, 2015) were succeeded by recurrent, multi-task, and physics-guided river-network architectures reporting large error reductions (Rahmani et al. (2021); Sadler et al. (2022); Barclay et al. (2023)) and by graph architectures making spatial dependence explicit (Topp et al., 2023).

Those gains are heterogeneous and the heterogeneity does not organize itself by architecture. Published accuracies are not comparable because reference models, key sets, and splitting conventions differ (Corona & Hogue, 2025), and where families *are* compared under a common protocol the spread is small: neither a comparison of statistical and machine-learning water-temperature models including an air2stream formulation (Feigl et al., 2021) nor one across recurrent, convolutional, and transformer architectures (Liu et al., 2025) finds an ordering that would explain the published dispersion. It survives inside a single panel: here, one fixed model on one fixed set of prediction days reports a seven-day skill of +0.251 or +0.038 depending only on which reference sits in the denominator, and no architectural account can produce that spread because the architecture does not change between the two numbers.

That a score is a statement about its benchmark is not contested, from the differential split-sample test (Kleměš, 1986) through explicit upper and lower benchmarks (Seibert et al., 2018) to benchmarking as the centre of what machine learning can tell us about a hydrological system (Nearing et al., 2021). What is missing for daily river temperature is the measurement, and three design choices make it attributable. Skill is usually quoted against naive persistence or climatology, but this variable has enough thermal inertia that relaxing the last observation toward a climatology already removes most of the error a learned model can remove. Models are scored on model-specific complete-case sets, so one can gain accuracy by declining to predict on hard days, and preprocessing is commonly fitted on windows including the evaluated interval (Arsenault et al., 2018). And spatial generalization is reported from random held-site splits, where a held gauge's neighbours remain in training (Kratzert et al., 2019). None is visible in a reported RMSE and each can inflate it.

Two boundaries follow, and recent work sits beyond both. Every model here consumes issue-time information only and is scored on retrospective gridded meteorology, not on what a forecaster could have run (Zwart, Oliver, et al. (2023); Padrón et al. (2025)); and every arm retains the target gauge's own observed history, so nothing here speaks to reaches that were never instrumented, now addressed at CONUS scale by frameworks built for that case (Bakar Siddik et al. (2026); Philippus et al. (2026)) and in far sparser networks (Chang et al., 2025). We measure how much a benchmark's construction moves a reported number, and neither of those.

We therefore hold the model and the data fixed and vary the design. The panel is 657,480 site-days from 120 stable USGS site numbers over 2006–2020, in 34 states and 15 two-digit hydrologic unit (HUC2) regions; every model is tuned on data through 2020 and scored on one common set of station/date/horizon keys at 1-, 3- and 7-day leads, with a held-out 2021–2023 window. The cohort is availability-selected rather than drawn, bounding what it can generalize to (Sections 3.4, 6). ThermoRoute, whose skill is decomposed, is described in Section 3.1; its architecture is the object under test, not the contribution.

2 Data and design

2.1 The frozen panel and the cohort it defines

The development record is a single derived panel bound by a manifest to a station registry (SI01). Each of 120 sites carries one row per calendar day from 2006-01-01 through 2020-12-31, giving 657,480 rows, missing observations being explicit nulls rather than absent rows. The models consume seven raw variables — water temperature, streamflow, air temperature, precipitation, a relative-humidity proxy, daylight-period mean shortwave radiation and wind speed — from USGS NWIS daily values (United States Geological Survey, 1994), Daymet V4 (Thornton et al., 2022) and gridMET (Abatzoglou, 2013).

The cohort was not sampled at random and nothing treats it as though it were: selection rejected 1,345 of 1,465 candidates, mostly for lacking joint water-temperature and discharge availability, and 2019–2020 participated in its construction. Water temperature is absent on 15.8% of rows and discharge on 2.8%; 38 retained stations share a hydrologic unit code with another and 19 lie within 10 km of another, so station-level independence is not tenable.

2.2 Temporal roles and the common key set

Temporal roles are fixed before any model is fitted, and a training sample is admitted only if its issue date and all target dates fall in the same partition: training 2006–2015 (438,240 rows, 341,646 with observed water temperature), validation 2016–2017, calibration 2018, development evaluation 2019–2020, and the held-out test 2021–2023 (per-partition counts in SI01). Models are tuned exclusively on data through 2020; the held-out window is used only for the comparative evaluation of Section 4.3.

The key registry is a data rule, not a model outcome. A key is admissible when its issue and target dates fall inside the window, issue-date water temperature is finite and observed, a 32-day history can be built, and target-date water temperature is finite and observed under the inclusive qualifier policy, evaluated independently at each lead. No condition mentions a model, and every declared model must produce a row on every admissible key: failing to predict is a failure,

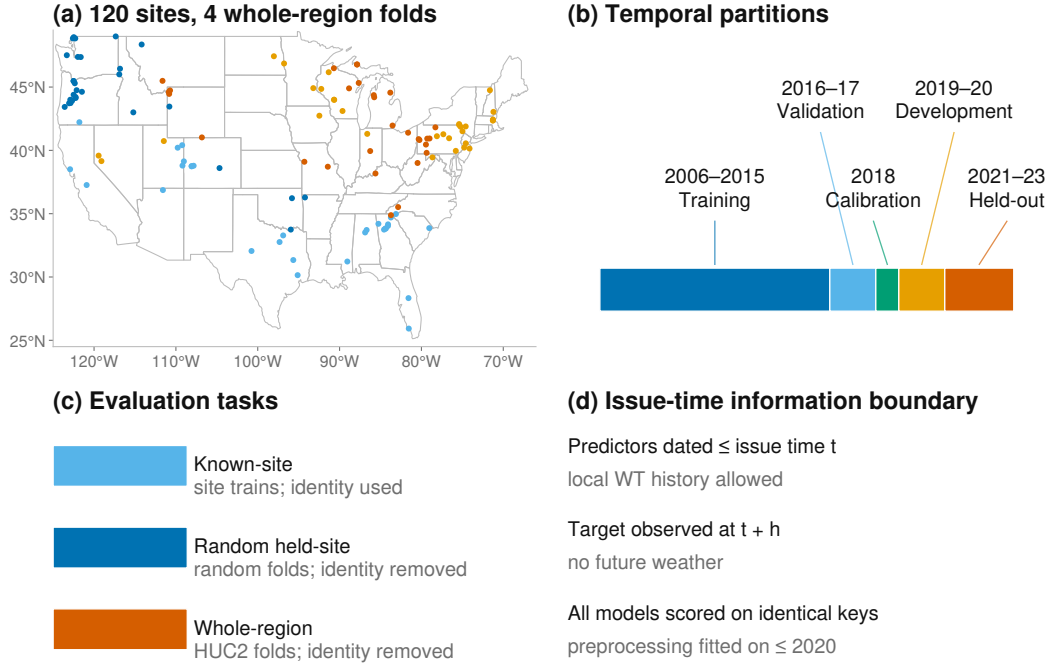


Figure 1. Study sites and evaluation design. (a) The 120 U.S. gauges coloured by the four deterministic whole-HUC2-region folds. (b) Temporal partitions. (c) The three evaluation tasks: known-site forecasting, random held-site transfer, and whole-region gauged transfer. (d) The issue-time information boundary — every model scored on the identical common forecast-key registry, no model-specific complete-case set permitted.

not an exclusion. The alternative — intersecting the days each model happened to score — is the second of the three inflation channels this study removes (Section 5.1) and would be indistinguishable from this in a reported RMSE. The development registry holds 249,072 keys, 83,024 per lead, and never carries a held-out claim.

2.3 Evaluation-period inputs and the information boundary

The test interval is 2021-01-01 through 2023-12-31 for the same cohort, with meteorology taken at each station coordinate rather than aggregated over upstream catchments — a real limitation for large basins (Section 6). The primary information set uses provider values dated no later than each issue date and consumes no horizon-specific forecast: a date-indexed retrospective hindcast, not a re-execution of what a forecaster could have run, because the as-issued provisional vintage of a gridded product cannot be reconstructed afterwards. Archiving requests, responses,

timestamps and checksums freezes the dataset evaluated without proving those values were available at the time.

Water temperature (NWIS 00010, °C), discharge (00060) and gage height (00065) come from NWIS as the provider's 00003 daily mean on site-local dates.

Discharge is served in $\text{ft}^3 \text{ s}^{-1}$ and consumed in those units: it enters the models only after per-station standardization, so the unit affects no reported quantity, but nothing converts it and the panel column is not $\text{m}^3 \text{ s}^{-1}$ (factors 0.0283168 and 0.3048). Gage height is provenance, not an input; every reported result is in °C. Values are retained irrespective of approval qualifier, which is preserved rather than used to select sites, dates or models; two consequences are disclosed rather than corrected, in that 2,059 of 657,480 rows (0.31%) carry negative discharge — a real measurement at tidal and regulated reaches — and observed water temperature spans -0.90 to 34.30 °C with 1,304 rows below -0.1 °C. The qualifier ledger, sensor-continuity limitation and provider-specific day definitions are in SI12. Test-window daily means are outcomes only: no model-, feature-, threshold-, calibration- or station-selection code reads them, all being fixed through 2020.

3 Methods

3.1 The predictor under test

ThermoRoute predicts at issue time t , station i , and lead h by adding a bounded learned residual to a damped-persistence anchor,

$$(1) A_{i,t+h} = c_{i,t+h} + \phi_i^h (y_{i,t} - c_{i,t})$$

where c is a seasonal climatology and ϕ_i a per-station decay fitted by no-intercept least squares on training-period consecutive-day anomaly pairs. The same object, fitted identically, is the damped-persistence reference.

The residual comes from a strictly left-looking temporal convolutional encoder over a 14-lag window, with a horizon-conditioned variable/lag router, a three-expert mixture, and a ± 1 °C algebraic bound on the deviation from the anchor. One model serves all three leads; 38,505 trainable parameters; five seeds averaged (Text S3,

Common anchor–residual formulation under matched information

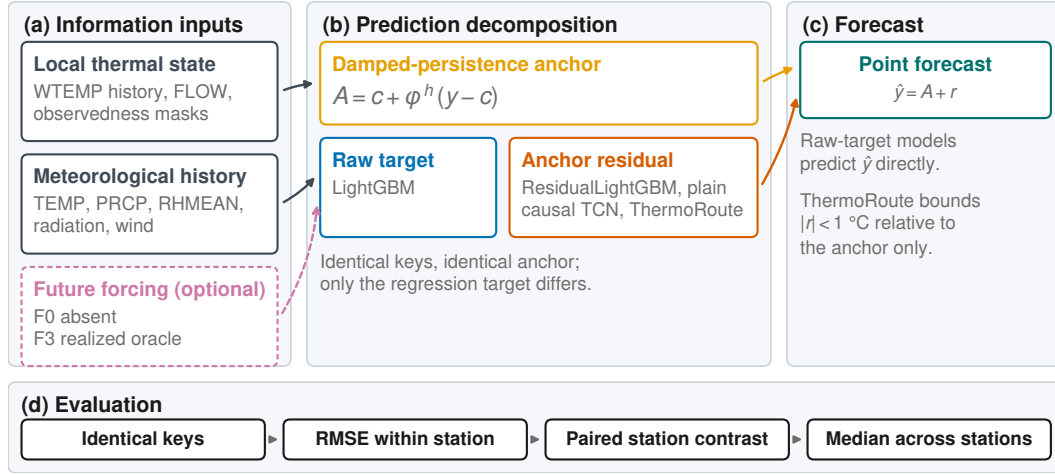


Figure 2. The formulation every compared model shares. LightGBM is a *raw-target* model with site identity; ResidualLightGBM, the plain causal TCN and ThermoRoute predict a residual around the same anchor, on identical keys and the same information set. The figure is structural and carries no evidence from either period; it is here because the fairness controls this paper is about are easier to check in a diagram than in prose.

Figure S11). The architecture is the object under test rather than the contribution: Section 4.2 shows an information-matched plain causal network reproduces it to within 0.023°C .

3.2 The reference set

The reference set is what the reported gain is a gain over, and each member removes one alternative explanation for apparent skill. **Persistence** ($\hat{y}_{\{t+h\}} = y_t$) quantifies raw series memory; **damped persistence**, equation (1) fitted on the training period, is the primary reference; **climatology** isolates the seasonal cycle. **LightGBM** (Ke et al., 2017) is the principal learned reference, run globally with site identity and per station. A **global LSTM** with a station embedding represents the deep sequence family (Rahmani et al. (2021); Zwart, Diaz, et al. (2023)). A **plain causal temporal convolutional network** is the information-matched deep reference: same outcome-free inputs, parameter count matched to within 2% (38,346 against 38,505), an unrestricted residual around the same anchor, and none of the relaxation proposal, router, mixture, or bound. An **air2stream-style hybrid**

(Toffolon and Piccolroaz, 2015) is fitted per station on the same keys (SI07); it is an unofficial empirical variant, so no process-side claim rests on it.

Tuning budgets are documented but *not* equalized across classes: any statement about LightGBM is scoped to this four-candidate procedure and feature schema, not to gradient boosting in general. The seven one-factor architecture controls are deletion sensitivities and do not prove component necessity.

3.3 Leakage control and spatial partitions

Leakage control is a set of mechanisms checkable against the code, not a statement of intent: every predictor carries a date no later than the issue date, the encoder is strictly left-looking, standardization constants, q90 event thresholds (2006–2015) and the 2018 conformal offsets and Platt calibrators are each fitted strictly before the interval they apply to, and a fresh interpreter replays every trained member rather than trusting self-consistent prediction files. One honest gap: the issue-date auxiliary quantities behind the relaxation proposal and the regime gate carry no separate mask, so the output is not guaranteed invariant to fill values.

Random held-site splits can be optimistic, because a held gauge's neighbours remain in training and the model reaches its thermal regime through correlated forcing and shared preprocessing. We therefore run a 2×2 factorial separating the spatial partition from the local-adaptation policy (protocol v1; SI11): the **random held-site** arm uses four balanced folds and is the one comparable to published random-split results, which is why it is not the spatial test, while the **held-region** arm packs the 15 HUC2 groups into four folds of [30, 30, 31, 29] stations at a mean 289 km from the nearest training gauge. *Local* adaptation lets a station's own history enter its climatology, anchor and imputation medians; *pooled* fits those over in-fold stations only. Held sites still supply their own water temperature through the issue date, so this is **gauged** transfer to a region whose gauges were not fitted, not prediction at a site with no record. Split-conformalized intervals are calibrated on 2018 only and reported as a marginal coverage diagnostic (SI08).

3.4 Estimand, metrics, and comparison set

Two differently signed quantities are reported and never combined:

the paired effect **(2)** $\Delta\text{RMSE} = \text{RMSE}(\text{cand}) - \text{RMSE}(\text{ref})$ in $^{\circ}\text{C}$, where **negative** favors the candidate, and the skill score **(3)** $\text{skill} = 1 - \text{RMSE}(\text{cand})/\text{RMSE}(\text{ref})$, dimensionless, where **positive** does. Equation (2) is the formal estimand and carries a $^{\circ}\text{C}$ unit; equation (3) is a bare signed number, and every value in Section 4 is labeled with which it is. The sampling unit is the station: unweighted RMSE is computed on the common daily keys per reportable station and the primary effect is the unweighted median across stations of equation (2) with ThermoRoute as candidate, a station/lead cell being reportable only with at least 100 valid paired targets.

The comparison set is five rows and they are not the same kind of test: three compare ThermoRoute with damped persistence at 1, 3, and 7 days as *superiority* tests at a 0.00 $^{\circ}\text{C}$ margin, two compare it with LightGBM at 3 and 7 days as *non-inferiority* tests at a +0.05 $^{\circ}\text{C}$ margin. That margin is a numerical ceiling on allowable degradation fixed before the outcomes were opened; it is not derived from sensor precision, biological response, or regulation, and nothing here is written as an equivalence or parity statement.

Two uncertainty procedures accompany each row, both clustered at HUC2: a one-sided p from exact enumeration of all 2^K whole-cluster sign vectors against that row's own margin, and a 10,000-draw cluster bootstrap percentile interval, Holm-adjusted (Holm, 1979) over the five. A row is supported only when both agree — Holm $p \leq 0.05$ *and* the whole interval below the margin. The cohort came from an availability filter rather than probability sampling of regions, so exchangeability is not established and the clustered intervals are approximate sensitivities, not decision evidence.

4 Results

Three evidence grades are used and they are not interchangeable.

Section 4.3 is *confirmatory*: suite, family, estimand, margins and decision rules were

fixed on data through 2020 and the window opened only to score them. Sections 4.1–4.2 are *development diagnostics* on the 2019–2020 partition, which participated in cohort construction, so they are the basis on which the suite was fixed rather than a test of it; their values are five-seed means on the 249,072 common development keys (SI20). Sections 4.4–4.6 are *post-outcome descriptive*, specified after the window had been opened, and no subsequent care converts a post-outcome specification into a confirmatory test.

Those descriptive results are nevertheless the largest effects this study measures and three are quoted in the Abstract, Key Points and Conclusions — a deliberate choice with a cost, so every quotation carries the label at the point of use and SI20 maps each headline number to its grade, artifact and specification date. They are hypotheses this cohort generated, not findings it confirmed. Every value below is an unweighted station median over reportable stations on the relevant common key registry, in °C.

4.1 The reference model, not the architecture, sets the reported gain

Most of a reported gain does not survive a strong reference. On identical keys the same fixed model reports a seven-day median station skill of +0.251 against persistence and +0.038 against damped persistence — a factor of 6.6 between two numbers differing only in the denominator of equation (3). The persistence column alone describes this model as gaining a quarter of the seven-day error budget; both columns describe the same fitted weights as gaining four percent (the full development-period comparison is Table S20.3).

The reference is itself a fitted object, and its construction moves the headline. Rebuilding the anchor under seven predeclared one-factor variants and rescored each against the *published* predictions, so every difference is attributable to the reference alone, moves seven-day skill from +0.025 to +0.063 with the published +0.038 in the middle (`outputs/final/anchor_sensitivity.parquet`); the low end is a decay fitted directly at each lead, a *stronger* reference. The number this paper quotes most often is the one most sensitive to how the reference was specified: the better the anchor, the less the learned model adds.

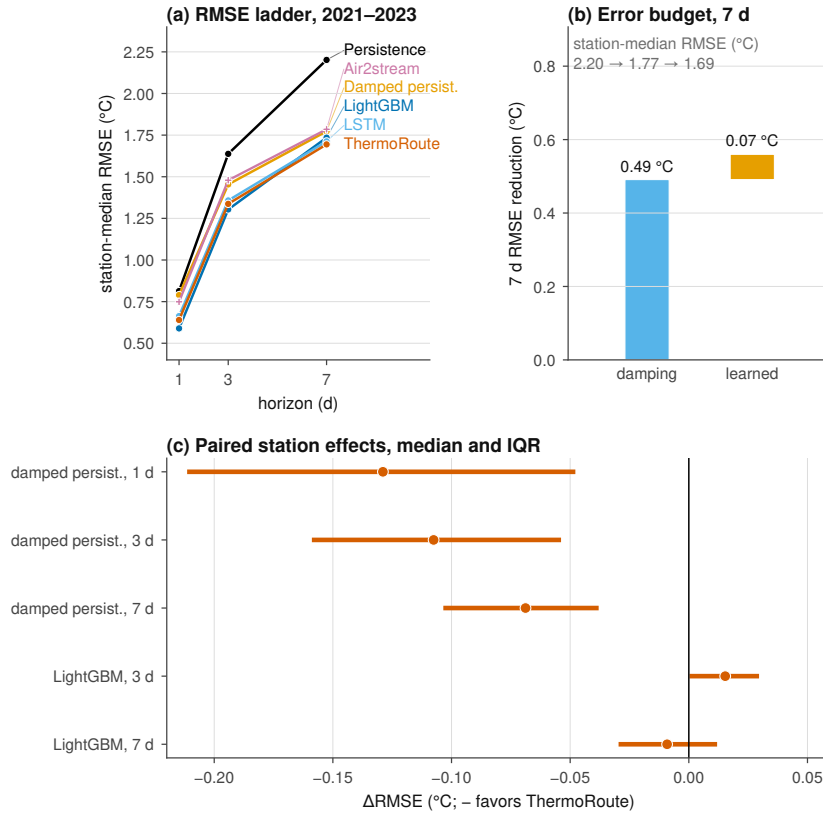


Figure 3. Decomposition of reported skill on the held-out 2021–2023 window (116 reportable stations). (a) Station-median RMSE by lead, every compared model. (b) The seven-day error budget: 0.49 °C of the reduction comes from damping the anchor and 0.07 °C from the learned residual. (c) Paired station effects for the five sealed tests, median and interquartile range; negative values favor ThermoRoute. Panel (c) shows the dispersion of the paired effects, not the narrower cluster-bootstrap intervals of Table 2.

4.2 A tree is the most accurate model, and the architecture is reproducible

On the development window the tree has the lowest station-median RMSE at all three leads (0.578, 1.280 and 1.649 °C against ThermoRoute's 0.631, 1.291 and 1.657 °C), with paired differences of +0.046, +0.008 and −0.005 °C and ThermoRoute win rates of 0.00, 0.40 and 0.60. The architecture's remaining argument is not accuracy but the ± 1 °C algebraic bound, which held on 100.00% of audited rows — a property of the construction, not a fitted outcome.

Very little of the architecture's structure is doing work either. Against the information-matched plain causal network (same inputs, anchor and optimiser bud-

get; 38,346 against 38,505 parameters) the full model's within-seed paired median ranges from -0.0003 to -0.0105 °C at 1 day and -0.0035 to -0.0226 °C at 7 days, every seed favouring the full model. The deletions agree: removing the temporal encoder is the largest single effect (1-day RMSE 0.631 to 0.679 °C), while removing the router, mixture, dynamic prior, fixed-relaxation constraint or residual bound moves it by at most 0.004 °C (SI09). These are deletion sensitivities and do not prove necessity.

4.3 Held-out 2021–2023 evaluation

The held-out evaluation applies the Section 3 suite and comparison set, fixed on data through 2020, to 2021–2023, on the common held-out registry ($n = 116$; per-model metrics in SI07). Station-median skill is $+0.250$ against persistence and $+0.038$ against damped persistence, the tree with site identity leads at one and three days (0.589 and 1.304 °C), and at seven days ThermoRoute reaches 1.694 °C against the tree's 1.735 °C. Against the information-matched plain convolutional network the paired station median is -0.015 °C at seven days in ThermoRoute's favour, a contrast outside the sealed family carrying no decision.

Each test is read at the margin it was sealed at (Section 3.4). The margins were fixed on 2026-07-22 under an attestation that no post-2020 outcome had been requested or inspected (`protocols/route_a_confirmatory_v1.json`, sealed by `route_a_protocol_seal_v1.json`), and the protocol's wording limit forbids reading the $+0.05$ °C ceiling as equivalence, parity or ecological negligibility. All five rows satisfy their sealed decision rule, with two limits: row 4 is a *supported non-inferiority* result at a lead where ThermoRoute is the worse model ($+0.015$ °C, interval excluding zero from above), and at seven days the median of -0.009 °C has an interval covering zero, so that row establishes the ceiling and not a win. And the seal is repository-internal Git and SHA-256 evidence, which the protocol calls sufficient for an honest owner and not proof against a repository owner rewriting history: a referee should read the confirmatory label as auditable rather than attested.

Table 1 — every model on the common held-out keys. Station-median RMSE in °C at each lead, 116 reportable stations, identical keys for every row; MAE and bias are the same rows in SI07.

Model	RMSE 1 d	RMSE 3 d	RMSE 7 d
Persistence	0.813	1.638	2.202
DampedPersistence	0.789	1.454	1.773
Climatology	1.899	1.902	1.903
LightGBM	0.589	1.304	1.735
LSTM	0.663	1.358	1.712
PlainMLP-7var	0.690	1.374	1.720
PlainCausalTCN-7var	0.646	1.330	1.710
Air2stream	0.719	1.459	1.825
ThermoRoute	0.640	1.337	1.694

Table 2 — the five sealed tests, each at its registered margin. ThermoRoute is the candidate and 116 stations are reportable throughout. Station-level Δ RMSE in $^{\circ}\text{C}$, negative favors ThermoRoute; whole-HUC2 cluster-bootstrap intervals; sign-flip p computed against the row's own margin, Holm-adjusted (unadjusted $p \leq 6.1 \times 10^{-5}$ throughout, the floor for 15 clusters). Rows 4 and 5 were registered as non-inferiority tests, so a margin of zero would misreport them (`outputs/final/sealed_confirmatory_family_v1/`).

#	Reference	Lead	Margin	Δ RMSE	95% CI	Win	Holm p	Sealed decision
			($^{\circ}\text{C}$)	($^{\circ}\text{C}$)		rate		
1	DampedPersistence	1 d	+0.00	-0.129	[-0.199, -0.076]	0.90	1.5e-04	SUPPORTED
2	DampedPersistence	3 d	+0.00	-0.108	[-0.143, -0.074]	0.91	1.5e-04	SUPPORTED
3	DampedPersistence	7 d	+0.00	-0.069	[-0.084, -0.057]	0.95	1.8e-04	SUPPORTED
4	LightGBM	3 d	+0.05	0.015	[0.010, 0.025]	0.24	1.8e-04	SUPPORTED
5	LightGBM	7 d	+0.05	-0.009	[-0.017, 0.003]	0.59	1.8e-04	SUPPORTED

One-factor ablations neither help nor hurt systematically: fixing the dynamic prior and removing the bounded-residual constraint give one-day skill against damped persistence of +0.180 and +0.176 against +0.173 for the full model (SI09). Neither adds material point accuracy on the independent window.

4.4 Matched spatial-transfer experiment on 2021–2023

The development-period whole-region finding (SI19) is re-tested on the independent window with a matched 2×2 factorial separating spatial geometry from local adaptation: station-agnostic trees under four leave-HUC2-region folds and four balanced random folds over five split seeds, crossed with local and pooled preprocessing (protocol v1; SI11, Figure S12). The design moves the median nearest-training-gauge distance from 60 to 263 km, and the region-minus-random paired penalty for the raw-target tree is +0.006, +0.009 and +0.007 °C.

This experiment does not resolve a geometry effect, and we do not report one. The penalty is the size of its own resampling noise — the median per-site standard deviation across five seeds is 0.004–0.007 °C — and four folds give four independent regional observations, so we report it as *below this design's resolution*. The pooled arm is separately defective: its preprocessing reused fold 0's statistics across folds 1–3, whose held-out stations lie inside fold 0's training set (DLOG-012), so no number from it reaches the Discussion or Conclusions.

4.5 Hydrologic conditions governing incremental skill

The remaining gain concentrates in specific states. Station thermal memory — the anomaly half-life of the train-fitted anchor, not an e-folding time — has a median of 6.9 days (IQR 5.0–11.9). At seven days the median memory gain is 0.49 °C against a learned gain of 0.07 °C (Figure 3b), the median per-station fraction of the reduction delivered by seasonal memory is 0.875, and the correlation between log half-life and learned gain is -0.40 at one day: longer-memory stations leave less to add. With thresholds fitted on 2006–2015 only, the seven-day median paired Δ RMSE over damped persistence is -0.069 °C on all keys, largest under low thermal anomaly (-0.229 °C) and rapid recent cooling (-0.170 °C) and -0.088 °C under rapid recent warming, with no issue-time state reversing the ordering. Strat-

ified by the *outcome* instead — which a manager cannot do at issue time — days that subsequently warm rapidly reach -0.30 °C, and the coldest target decile is the one stratum where the anchor alone is marginally better ($+0.03$ °C).

The full sixteen-stratum table is in SI11 on two grounds its own numbers supply: the 100-key reportability rule qualifies a station on its *all-keys* record while the strata score it on a median of 71, and the strata are not a multiplicity family.

4.6 What the information is worth, and where the estimator starts to matter

Sections 4.4 and 4.5 leave one question open: the residual learned gain is small, but is that a limit of the models or of what they were given? Four crossed axes answer it, and **every result here is post-outcome and descriptive**. **L** withholds the target site's own observations (L0 keeps everything, L2 removes every target-site water-temperature input); **F** substitutes realized meteorology over $t+1 \dots t+h$ for the issue-time forcing, an oracle bound rather than a forecast product; **G** is the holdout geometry; **A** swaps the tree for the information-matched network. Prohibited inputs are dropped rather than filled, all 24 mask-invariance proofs return exactly zero, and a negative control re-admitting one water-temperature lag is caught with a 147 °C shift (levels, estimands, sealing and per-cell tables in SI20).

Local thermal state dominates. Withholding it costs 1.3–1.7 °C at every lead and at all 116 stations — two to thirteen times realized future meteorology and roughly seventy times the architecture effect of Section 4.2 — and almost all of that is the recent sequence rather than the long record (0.01–0.16 °C to pool a station's climatology and damped rate, 1.17–1.71 °C to remove its recent readings). Neither ablation tests prediction at a site that was never instrumented: a station with no thermal record has no target to score against (Section 6). The geometry penalty Section 4.4 could not resolve behaves the same way, 0.006–0.014 °C at L0 against 0.073–0.129 °C at L2, every double-difference interval excluding zero.

Realized future meteorology is worth 0.13 °C at one day and 0.54–0.63 °C at three and seven (Table 3), three to four times each cell's minimum detectable effect. It is event-scale weather information and not seasonal phase — a placebo sealed before the outcome existed retains 0.0–8.8% of the value, a second

sealed control displacing the future by a week retains -0.1 – 14.0% — and it is almost entirely air temperature (0.595 of the 0.627°C seven-day value), of which 49–62% survives an archived fixed-lead composite.

Table 3 — value of realized future meteorology. Station-first forcing value $V_F = \text{median}_i[\text{RMSE}_i(F0) - \text{RMSE}_i(F3)]$ in $^\circ\text{C}$; CI is a 10,000-draw whole-HUC2 cluster bootstrap and MDE the smallest effect this cohort's cluster structure resolves at $\alpha = 0.05$.

Model	Lead	RMSE F0	RMSE F3	V_F	95% CI	MDE
LightGBM	1 d	0.614	0.451	0.130	[0.081, 0.189]	0.031
LightGBM	3 d	1.330	0.765	0.542	[0.415, 0.721]	0.152
LightGBM	7 d	1.742	1.063	0.627	[0.423, 0.803]	0.172
ResidualLightGBM	1 d	0.606	0.446	0.125	[0.084, 0.183]	0.028
ResidualLightGBM	3 d	1.327	0.752	0.578	[0.406, 0.704]	0.160
ResidualLightGBM	7 d	1.722	1.087	0.605	[0.445, 0.795]	0.151

Crossed with L, forcing is worth *less* once local observations are withheld, not more (Table 4). Weather earns its value by correcting a trajectory, and at short leads that trajectory is the anchor built from the station's own readings, so the two are complements.

Table 4 — value of realized future meteorology by information level. Station-first paired values in $^\circ\text{C}$ under whole-region holdout, 116 reportable stations, 10,000-draw whole-HUC2 cluster bootstrap.

Forcing value	1 d	3 d	7 d
At L0, gauged	0.116 [0.059, 0.174]	0.474 [0.299, 0.693]	0.580 [0.366, 0.779]
At L2, obs. withheld	0.038 [0.020, 0.072]	0.331 [0.249, 0.496]	0.596 [0.431, 0.804]
Interaction, L2 – L0	–0.076 [–0.098, –0.056]	–0.147 [–0.193, –0.079]	+0.016 [–0.035, +0.099]

The estimator matters only where the information does not. At L0 the tree and the network differ by at most 0.009°C at any lead, against station-

median errors of 0.53–1.74 °C, under both geometries — a statement about two fitted objects and not two model families, since their tuning budgets were documented rather than equalized (Section 3.2) and no graph, recurrent or differentiable-hybrid model is represented. Remove the local gauge and the tree wins by 0.02–0.19 °C, and refitting those cells at a 300-epoch cap rather than the shared 40 makes the penalty *grow*, so the reported figures are the conservative pair. Given the oracle the sign reverses, to -0.096 $[-0.177, -0.038]$ at seven days: an argument for sequence models *conditional on good forcing*, not for the benchmarked regime. Both triple differences sit below their cells' minimum detectable effects, so they are underpowered rather than evidence of absence — and the sealed contract records that failure to reject is not equivalence.

5 Discussion

5.1 Why these numbers are smaller than published gains

The difference is one of design, not model quality. The reference: a study quoting skill against persistence or climatology reports a number comparable to our $+0.203$ to $+0.251$ column and not to our $+0.038$ to $+0.168$ column, differing by up to a factor of 6.6 for the same fitted weights, so we suggest damped persistence as the minimum reference for this variable. The spatial partition, where our design could not measure what it set out to (Section 4.3). And the key set and preprocessing boundary: scoring each model on its own complete-case set, or fitting statistics on windows including the evaluated interval, moves error downward without a trace. We do not assert the published literature is affected; removing both channels here left the constrained deep model behind a tree at 1 and 3 days.

5.2 What a benchmark of this geometry can carry, and what transfers

The evaluation design is stronger than its inferential reach. Clustered inference rests on resampling whole spatial units, and its accuracy depends on having enough units, of comparable size, that the asymptotics are not a fiction: fifteen HUC2 groups with an inverse-Herfindahl effective count of 9.54 and a largest group holding 21.7% of stations is not enough, and the design was never exchangeable

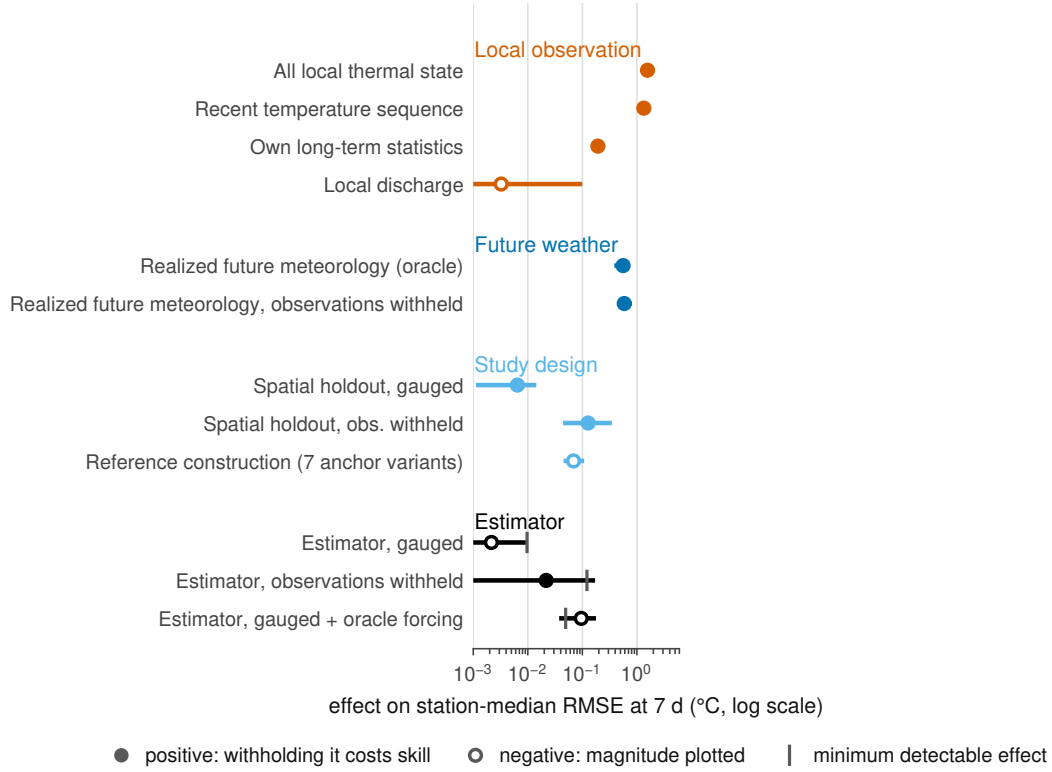


Figure 4. What each axis of the study is worth, at seven days. Station-first paired effects in °C on a log axis, damped-residual tree throughout; open markers are negative effects plotted at their magnitude, and the tick marks the minimum detectable effect where one is defined. Local observation, future weather, study design and the estimator span three orders of magnitude. The axes are separate conditional designs and are compared, never summed. Every row is post-outcome and descriptive except the last, and every bar is a cluster-bootstrap interval except the last: *reference construction* is the range across the seven predeclared anchor variants of Section 4.1 — a spread over specifications, not a sampling interval. It is on this axis because every other row is conditional on one anchor while the anchor is itself a fitted object, and it says how much of the figure is a property of the reference rather than of the river.

across regions. Usual practice would report a station-level interval and nothing in such a paper would tell a reader the effective cluster count is under ten; the difference is disclosure, not the data and not the model.

The study is narrow by construction — a common-key clustered benchmark of point predictors at gauged sites, not an attempt to replace process-based thermal models, river-network graph models or differentiable hybrid formulations (Jia et al. (2021); Rahmani et al. (2023); Zwart, Diaz, et al. (2023)). What is portable is not

the architecture but three controls: one key registry across models, statistics fitted strictly backwards in time, and partitions by whole hydrologic region.

6 Limitations

Scope. This is a descriptive benchmark on a fixed cohort and no interval or ranking here is decision evidence (Section 5.2). Section 4.6 withholds a station's own record from the model, which is not predicting where none was ever instrumented: cohort, climatology and evaluation keys all come from gauged sites, and a site with no record supplies no target to score against. Genuinely ungauged prediction is not quantified and no operational-replay claim is made; throughout, "causal" describes time ordering only.

Cohort and measurement. The cohort trades a modest discharge channel (removal costs $+0.042$ °C at 1 day; SI19) against wide spatial coverage, excluding 950 of 1,465 candidates for missing joint flow; meteorology is point-scale rather than basin-integrated; and the air2stream-style hybrid is an unofficial variant (Toffolon and Piccolroaz, 2015), so no process-side claim rests on its scores.

Reference and thresholds. The damped anchor is a fitted object and the seven-day headline moves by a factor of 2.5 across defensible constructions (Section 4.1), so every number against damped persistence is conditional on the anchor of Section 3.1 — primary because it was fixed before the window opened, not because it is strongest. The ± 1 °C bound, event quantiles and 100-key rule are development-selected, and q90 is a tail diagnostic with no biological or regulatory meaning (SI19).

7 Conclusions

We evaluated daily river water-temperature prediction at 120 stable U.S. gauges across 15 hydrologic regions on one common set of station/date/horizon keys, all statistics fitted strictly backwards in time, on a held-out 2021–2023 window. Three findings survive. The reference model governs the reported gain: the deep predictor's seven-day station-median skill is $+0.250$ against persistence but $+0.038$ against damped persistence. Model ranking does not favor architectural constraint:

a gradient-boosted tree with site identity has the lowest station-median RMSE at 1 and 3 days (0.589 and 1.304 °C), and both LightGBM rows satisfy their sealed +0.05 °C non-inferiority bound — a pre-declared numerical ceiling, not equivalence or parity — with the tree the better model at three days by +0.015 °C while staying under it. And the spatial-partition question is unresolvable in the regime it was posed in and resolvable just outside it: with the held station's own history retained, whole-region holdout changes station-median RMSE by less than the five-seed re-sampling spread, while withholding that history separates the same contrast (Section 4.6). The geometry of a split cannot be assessed independently of how much local information the model keeps.

Every model here is issue-time-only, and a post-outcome descriptive analysis bounds what perfect future meteorology could add at 0.13 °C at one day and 0.54–0.63 °C at three and seven, of which a sealed placebo removes 91–100%: the gain is event-scale weather information, not seasonal phase.

A published skill score is not interpretable without its reference model, its spatial partition and its information set, and every comparison here is descriptive for this fixed, availability-selected cohort.

Open Research Statement

Data. The derived panel, the station registry, their manifests and the test-window acquisition record are deposited as one versioned dataset at [DATA DOI TO BE MINTED] under [DATA LICENCE TO BE ASSIGNED]. Neither can be assigned before the byte-level rights review completes, because a derived product does not inherit the most permissive of its upstream terms; the deposit is planned and specified, not existing, and where redistribution terms are unresolved it carries the request specification and a SHA-256 manifest instead of the bytes (SI16). Primary sources are USGS NWIS (United States Geological Survey, 1994), Daymet V4 R1 (Thornton et al., 2022) and gridMET (Abatzoglou, 2013), each under its own terms.

Software and reproduction. The analysis software, per-stage manifests, and Python 3.12 dependency lock with package hashes are archived at [SOFTWARE DOI TO BE MINTED]; the MIT licence covers neither the observational data nor the archived provider responses. Sections 4.1–4.2 reproduce from the archived panel,

bundles and pinned environment and Section 4.3 from the acquisition record, with bit-level equality expected for the trees and a documented tolerance for the neural members (SI15). Reproduction has not been carried out by an operator independent of the authors on an independent host, and this statement is not a claim that it has.

Acknowledgments

[FUNDING, COMPUTATIONAL RESOURCES, COMPETING INTERESTS AND CREDIT ROLES TO BE COMPLETED — each funder with its award number or an explicit statement of none; a competing-interests declaration from every author; the signed intake form is docs/FAIR.SUBMISSION.READINESS_AND_TEMPLATES.md §2.]

We acknowledge the U.S. Geological Survey, the ORNL DAAC and the Northwest Knowledge Network for the products named in Open Research; acknowledgment of a provider is not an endorsement of this analysis.

Supporting Information

SI01 cohort and registry with the candidate-rejection ledger; SI02 issue-time boundary; SI03 equations and units; SI04 protocol, chronology and decision log; SI05–SI06 the comparison set; SI07 all-model scores on the exact common keys, with MAE and bias, including the air2stream-style hybrid; SI08 probability and reliability schemas; SI09 architecture and information-matched controls; SI10 temporal coverage; SI11 the matched spatial factorial, cluster geometry and the hydrologic-state strata; SI12 outcome quality control and qualifiers; SI13 the history-dependent external arm; SI14 missingness and failure cases; SI15 reproduction hashes and environment parity; SI16 rights and data dictionary; SI19 development-period spatial analyses; SI20 the method, estimands, sealing, evidence-grade table and relocated objects of the Section 4.6 arms.

Figures S1–S3 give cohort geometry, temporal roles and model dataflow; S4–S8 expand Sections 4.3 and 4.5; S9 is a development-period conformal sensitivity not comparable with any held-out figure; S10 the architecture controls; S12 the matched spatial-transfer panels, relocated because they show a different model from the section that cites them and an effect the design cannot resolve.

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